Data-Backed Journal-Style Replication of *Beyond Local Search*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 23,844 observations across 184 entities, and the batch headline estimate is -0.0389 for Boundary Existing Field vs Local with p = 0.39. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: boundary-spanning search; exploration; inventive impact; patent classes; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks whether boundary-spanning search improves inventive impact relative

to local search (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because boundary-spanning search claims are often treated broadly, while the original paper distinguishes among qualitatively different forms of exploratory search. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The executable evidence base used in this manuscript contains 23,844 observations across 184 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper sits in a top-journal literature on local search, boundary-spanning exploration, and the inventive consequences of moving beyond familiar domains. (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the replication uses the current local dataset to ask whether the most novel form of boundary-spanning search still outperforms local search when the taxonomy is rebuilt transparently from processed patent variables. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and

patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). Because this paper is about boundary-spanning inventive search, the literature review also adds classic patent-indicator economics and adjacent top-journal studies on temporal exploration, search-guiding science, organizational search balance, and alliance-based exploration so that the boundary-spanning mechanism is placed inside a broader evidence base rather than a narrow single-paper dialogue. (Griliches, 1990; Jaffe, 1989; Fleming and Sorenson, 2004). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Nerkar, 2003; Rivkin and Siggelkow, 2003). A final topic-specific literature block keeps the bibliography tied to the paper's actual mechanism rather than to a generic replication template. (Lavie et al., 2011; Ghosh et al., 2014).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Rosenkopf and Nerkar, 2001). The original study can be summarized as a mechanism paper with a sharp empirical core. The source paper contrasts local search with boundary-spanning exploration. Impact rather than patenting volume is the main empirical outcome. Industry-specific coding is part of the original design. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published

article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Rosenkopf and Nerkar (2001) starts with the published empirical design rather than with the local rerun. The source article distinguishes local search from two forms of boundary-spanning exploration and evaluates how those search modes shape patent impact in an industry-specific archival setting (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published study is a patent-based archival analysis with industry-specific coding of exploratory search. The published measurement architecture is described as follows: The published design distinguishes local search from boundary-spanning search and emphasizes impact rather than raw patent volume. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: patent-level impact models that compare boundary-spanning search modes with local search while preserving an industry-specific exploratory-search taxonomy. The key published comparison asks whether the more exploratory search modes, especially exploration into genuinely new fields, outperform local search in terms of inventive impact rather than patenting volume. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). This paper is a methodology-

specific local analogue because the original optical-disk-industry coding cannot be replayed from the current processed-data stack. The local rerun preserves the search taxonomy and impact-focused comparison, but it reconstructs the taxonomy from general experience counts instead of industry-specific manual coding (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source paper emphasizes different kinds of boundary-spanning search. The local reconstruction therefore adopts the following bounded decision: The local panel reconstructs a three-way search taxonomy from experience counts. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Impact is the key outcome. The local reconstruction therefore adopts the following bounded decision: The outcome remains impact-focused through citations and a top-tail indicator. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Industry-specific coding is part of the original design. The local reconstruction therefore adopts the following bounded decision: The industry-specific coding is generalized to the processed multi-firm patent panel. This bridge is

treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Patent-level recombination panel built from the cleaned reproduction dataset. Search type reconstructed from prior organizational and field experience counts. Impact outcomes measured with patent-level inventive impact and a top-5-percent indicator (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 23,844 observations across 184 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. The local sample frame is Patent-level recombination panel derived from the cleaned local reproduction dataset The observed time window in the executable

panel is 1986 to 2016 in the local executable panel The outcome is Patent-level inventive impact and a yearly top-tail impact indicator (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Rosenkopf and Nerkar, 2001; Hall et al., 2005).

3.1 Local Data Sources Used in This Replication

This replication uses the cleaned patent-level recombination panel, a search-type classification rebuilt from organizational and field experience counts, and patent-level inventive-impact and top-tail outcome variables already present in the local workflow (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Those local dataset families are not interchangeable: cleaned patent-level recombination panel is used to define the patent sample and breadth-degree controls; search-type classification from experience counts is used to classify patents as local, boundary-existing-field, or boundary-new-field search; patent-level impact and top-tail outcomes is used to measure inventive impact and top-tail impact. Together, they support 23,844 observations across 184 entities over 1986 to 2016 in the local executable panel (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. Patent-level recombination panel built from the cleaned reproduction dataset. Search type reconstructed from prior organizational and field experience counts. Impact outcomes measured with patent-level inventive impact and a top-5-percent indicator. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed dataset families and calculated outputs supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed through the body text (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

DatasetFamily	RoleInPaper
Cleaned patent-level recombination panel	define the patent sample and breadth-degree controls
Search-type classification from experience counts	classify patents as local, boundary-existing-field, or boundary-new-field search

DatasetFamily	RoleInPaper
Patent-level impact and top-tail outcomes	measure inventive impact and top-tail impact

Local data families used in the present paper.

3.2 Sample Reconstruction

The search-type file classifies patents into local, boundary-existing-field, and boundary-new-field search. Local patents account for 12,131 observations with mean impact 5.136. Boundary-existing-field patents account for 3,270 observations with mean impact 4.535. Boundary-new-field patents account for 8,443 observations with mean impact 5.860 (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Even before estimation, the raw means point to a substantive ranking in which the most novel form of search outperforms the more moderate boundary-spanning category. The top-tail shares follow the same direction: 4.11% for local search, 2.87% for boundary-existing-field search, and 4.58% for boundary-new-field search (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8,443	5.860	4.58%
local	12,131	5.136	4.11%
boundary_existing_field	3,270	4.535	2.87%

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Local search is operationalized locally as Prior organizational experience and appears in the executable dataset as search_type = local. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with positive prior organizational experience as local.. This mapping is retained because it preserves the construct's functional

role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Boundary-existing-field search is operationalized locally as No prior organizational but prior field experience and appears in the executable dataset as search_type = boundary_existing_field. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero prior organizational but positive field experience as boundary-existing-field.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Boundary-new-field search is operationalized locally as No prior organizational and no prior field experience and appears in the executable dataset as search_type = boundary_new_field. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja,

2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Inventive impact is operationalized locally as Patent-level citation impact and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local patent-level impact variable from the cleaned panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Tail impact is operationalized locally as Top-5-percent yearly impact indicator and appears in the executable dataset as high_top5_y. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the already materialized yearly top-5-percent indicator.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Local search	Prior organizational experience	Classify patents with positive prior organizational experience as local.
Boundary-existing-field search	No prior organizational but prior field experience	Classify patents with zero prior organizational but positive field experience as boundary-existing-field.

Construct	Local measure	Derivation
Boundary-new-field search	No prior organizational and no prior field experience	Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.
Inventive impact	Patent-level citation impact	Use the local patent-level impact variable from the cleaned panel.
Tail impact	Top-5-percent yearly impact indicator	Use the already materialized yearly top-5-percent indicator.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Rosenkopf and Nerkar (2001). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Local search, the manuscript uses Prior organizational experience under the exact local implementation search_type = local. The derivation step is: Classify patents with positive prior organizational experience as local.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Boundary-existing-field search, the manuscript uses No prior organizational but prior field experience under the exact local implementation search_type = boundary_existing_field. The derivation step is: Classify patents with zero prior organizational but positive field experience as boundary-existing-field.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Boundary-new-field search, the manuscript uses No prior organizational and no prior field experience under the exact local implementation search_type = boundary_new_field. The derivation step is: Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Inventive impact, the manuscript uses Patent-level ci-

tation impact under the exact local implementation inventive_impact. The derivation step is: Use the local patent-level impact variable from the cleaned panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Tail impact, the manuscript uses Top-5-percent yearly impact indicator under the exact local implementation high_top5_y. The derivation step is: Use the already materialized yearly top-5-percent indicator.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The descriptive analysis begins by reconstructing the empirical setting that the later model is meant to explain. For Inventive impact, the mean is 5.310, the standard deviation is 17.55, and the interquartile range runs from 1.000 to 6.000; For Top-5 indicator, the mean is 0.0411, the standard deviation is 0.199, and the interquartile range runs from 0.00000 to 0.00000; For Breadth of recombination, the mean is 4.294, the standard deviation is 4.567, and the interquartile range runs from 2.000 to 5.000; For Recombination degree, the mean is 0.555, the standard deviation is 0.174, and the interquartile range runs from 0.444 to 0.691 . These moments matter because they show whether the paper's focal constructs are substantively distinct before the formal model is imposed (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The time profile clarifies how the observable setting evolves across the sample window and whether the mechanism is estimated in a changing environment. the value of patents is 643 in 1986 and 534 in 2016; the value of mean impact is 4.020 in 1986 and 3.768 in 2016; the value of mean breadth is 2.869 in 1986 and 4.629 in 2016 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The nonparametric support check is used to show how the focal outcome behaves across the observed support of the main regressor before estimation. Boundary-new-field search has mean impact 5.860 and top-tail share

4.58%, local search has mean impact 5.136 and top-tail share 4.11%, and boundary-existing-field search has mean impact 4.535 and top-tail share 2.87% (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The correlation structure is reported because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002).

4.1 Setting-Specific Figure Evidence

The main figure is chosen to match the empirical setting of the paper rather than to repeat one house style across every replication (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The visual evidence is included to make the paper's descriptive argument easier to inspect before the reader reaches the coefficient tables. The grouped figure makes the search ranking directly visible: local search has mean impact 5.136, boundary-existing-field search has mean impact 4.535, and boundary-new-field search has mean impact 5.860. That visual ordering matters because the paper's main claims depend on whether more distant search is already advantaged in raw outcome space (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002).

4.2 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. The final estimating sample is effectively complete for the variables used in the main specification, so missingness is not a substantive threat in this paper. For that reason, the manuscript does not spend body-table space on an all-zero missingness display (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather

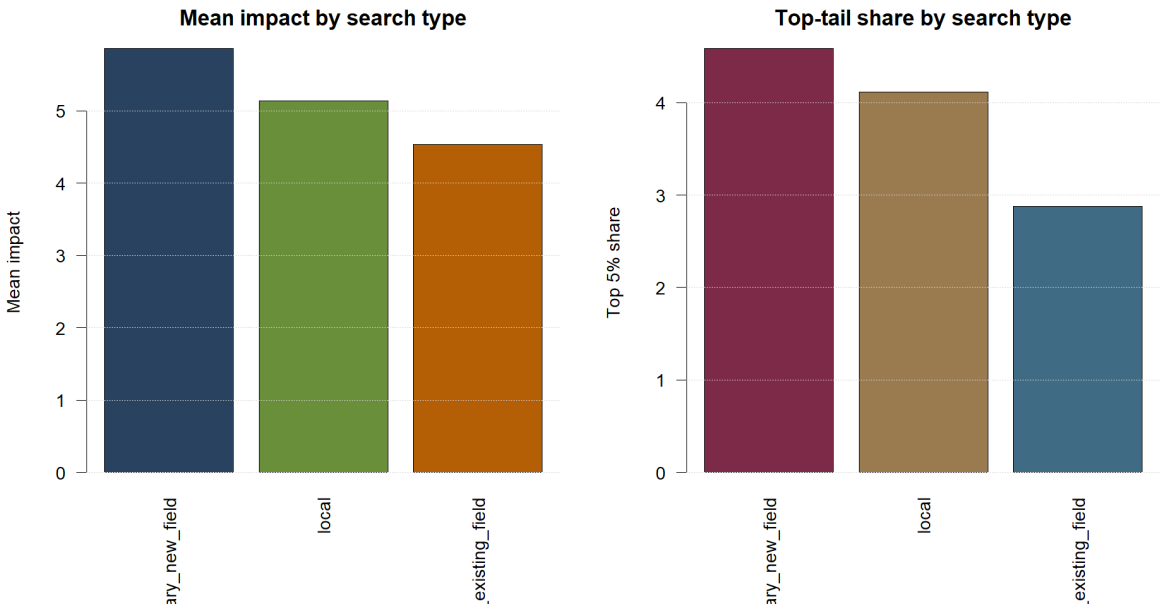

Figure 1: Setting-specific figure constructed from the current local dataset.

than assumed. The most heavily represented entities contribute between 830 and 3,200 observations each. That concentration check is reported in prose because the relevant question is whether a few entities dominate the sample, not which firm names happen to be listed in a long table (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 2.411 and the mean outcome is 4.464 across 13,675 observations. In the highest observed quantile, the corresponding values are 9.963, 8.115, and 4,438 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because boundary-spanning search claims are often treated broadly, while the original paper distinguishes among qualitatively different forms of exploratory search. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	23,844	5.310	17.55	1.000	2.000	6.000	0	812
Top-5 indicator	23,844	0.0411	0.199	0.0000	0.0000	0.0000	0	1
Breadth of recombination	23,844	4.294	4.567	2.000	3.000	5.000	2	117

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Recombination degree	23,844	0.555	0.174	0.444	0.560	0.691	0	1
Field experience	23,844	1.634	1.820	0.0000	1.099	2.773	0	8
Organizational experience	23,844	0.923	1.317	0.0000	0.693	1.386	0	8
Prior-art citations	23,844	15.09	50.45	5.000	8.000	13.00	2	2263
Same-year patents	23,844	728.7	1,303	109.0	313.0	652.0	1	7321
Prior cumulative stock	23,844	7.507	2.214	6.330	7.693	8.893	0	11

Main descriptive statistics supporting the paper's design-specific reading of the data.

Period	Patents	MeanImpact	MeanBreadth	MeanDegree	NewFieldShare
1986	643	4.020	2.869	0.517	42.77%
1987	789	4.280	2.918	0.518	43.35%
1988	1,066	4.461	3.132	0.532	43.53%
1989	1,390	3.934	3.081	0.527	40.07%
1990	1,349	4.514	3.116	0.522	37.44%
1991	1,325	4.738	3.150	0.531	37.43%
1992	1,302	5.015	3.332	0.535	40.40%
1993	1,354	5.672	3.461	0.532	38.77%
1994	1,330	6.265	3.578	0.541	39.32%
1995	1,217	6.260	3.844	0.556	37.80%
1996	659	6.727	3.838	0.531	37.63%
1997	646	6.014	4.245	0.552	41.95%
1998	356	6.295	3.924	0.547	40.45%
1999	654	5.483	3.810	0.544	37.00%
2000	636	5.541	4.022	0.540	36.32%
2001	614	5.151	4.414	0.562	41.69%
2002	632	4.889	4.384	0.558	36.87%
2003	522	4.502	4.418	0.550	34.87%
2004	324	3.664	4.809	0.573	41.36%
2005	284	4.500	5.486	0.576	40.85%

Period	Patents	MeanImpact	MeanBreadth	MeanDegree	NewFieldShare
2006	336	4.890	5.265	0.566	32.14%
2007	246	3.654	5.268	0.569	30.89%
2008	535	3.652	4.512	0.590	36.07%
2009	206	4.801	6.005	0.599	29.61%
2010	511	4.417	6.648	0.608	25.64%
2011	651	6.452	6.275	0.609	25.50%
2012	1,068	4.000	6.519	0.611	26.03%
2013	868	5.386	6.730	0.598	26.04%
2014	725	12.64	6.989	0.600	22.90%
2015	1,072	6.616	6.174	0.597	20.43%
2016	534	3.768	4.629	0.581	16.67%

Observed time patterns relevant to the focal empirical setting.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8,443	5.860	4.58%
local	12,131	5.136	4.11%
boundary_existing_field	3,270	4.535	2.87%

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Inventive impact	Top-5 indicator	Breadth of recombination	Recombination degree	Field experience	Organizational experience	Prior-art citations	Same-year patents	Prior cumulative stock
Inventive impact	1.000	0.455	0.364	0.018	-0.021	-0.008	0.591	0	0.022
Top-5 indicator	0.455	1.000	0.129	0.024	-0.012	0.006	0.149	0	0.005
Breadth of recombination	0.364	0.129	1.000	0.466	-0.263	-0.144	0.822	0	0.149
Recombination degree	0.018	0.024	0.466	1.000	-0.456	-0.313	0.140	0	0.122
Field experience	-0.021	-0.012	-0.263	-0.456	1.000	0.838	-0.077	0	0.132
Organizational experience	-0.008	0.006	-0.144	-0.313	0.838	1.000	-0.018	0	0.267

Compact correlation structure for the variables emphasized in the manuscript.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is Patent-level impact and top-tail models with search-type indicators and patent-level controls. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Rosenkopf and Nerkar, 2001; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002).

5.1 Design Decisions and Local Substitutions

The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel derived from the cleaned local reproduction dataset, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact and a yearly top-tail impact indicator, and the main explanatory design is Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The third

set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects.. The crucial exclusion or substitution rule is: The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel derived from the cleaned local reproduction dataset over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact and a yearly top-tail impact indicator and the main explanatory design is Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects. The main exclusion or substitution rule is The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

5.2 Estimating Equation

$$E[\text{Impact}_{pft} \mid X_{pft}] = \exp\left(\beta_1 \mathbb{1}[\text{BoundaryExisting}_p] + \beta_2 \mathbb{1}[\text{BoundaryNew}_p] + \beta_3 X_{pft} + \alpha_f + \gamma_t\right). \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.0389 on Boundary Existing Field vs Local (p = 0.39). The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. Boundary new field versus local is estimated at 0.113 with p = 0.001. Recombination degree is estimated at -0.411 with p = 0.023. Prior-art citations is estimated at 0.0013 with p = 0.055 (Rosenkopf and Nerkar, 2001; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Model	Term	Estimate	Std. error	P-value
Rosenkopf Impact	Boundary existing field versus local	-0.0389	0.0449	0.387
Rosenkopf Impact	Boundary new field versus local	0.1126	0.0353	0.0014
Rosenkopf Impact	Breadth of recombination	0.0223	0.0130	0.0870
Rosenkopf Impact	Recombination degree	-0.4108	0.1802	0.0226
Rosenkopf Impact	Prior-art citations	0.0013	0.0007	0.0548
Rosenkopf Impact	Log prior patent stock	-0.1965	0.1067	0.0657

Key coefficient table used in the main text.

6.2 Visual Summary of the Main Estimates

The coefficient figure is included to show the relative magnitude and uncertainty of the main terms without forcing the reader to reconstruct that pattern from columns of numbers alone (Rosenkopf and Nerkar, 2001; Hall et al., 2005).

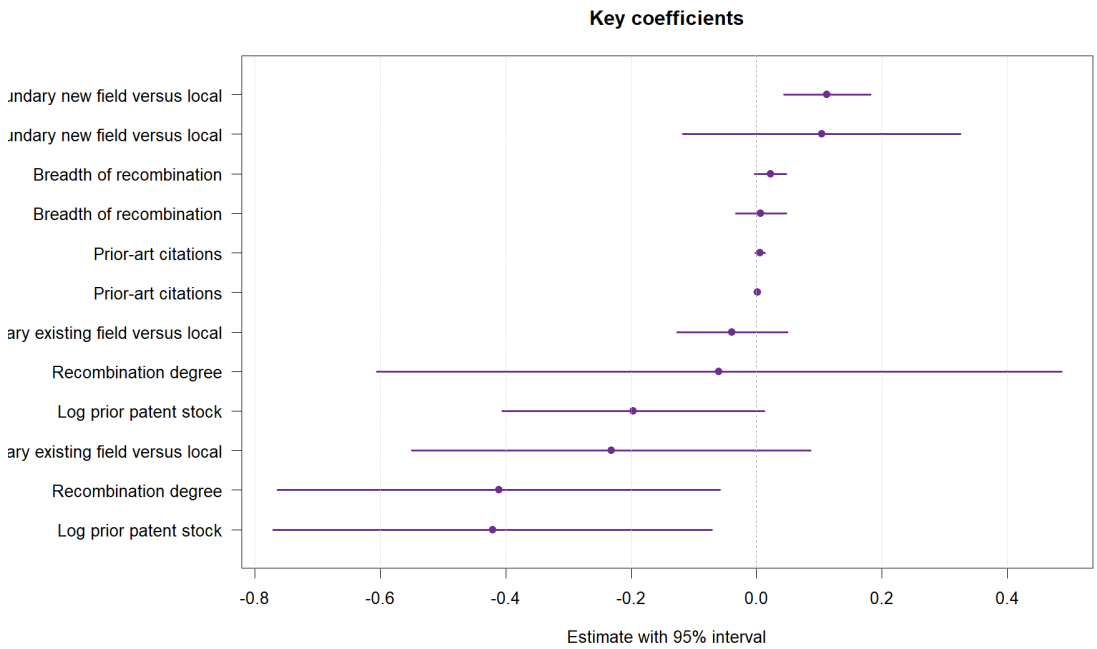

Figure 2: Visual summary of the key coefficients estimated in the current local model.

6.3 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Boundary new field versus local is estimated at 0.113 with a p-value of 0.001; Recombination degree is estimated at -0.411 with a p-value of 0.023; Prior-art citations is estimated at 0.0013 with a p-value of 0.055; Log prior patent stock is estimated at -0.196 with a p-value of 0.066 (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Boundary new field versus local, which is estimated at 0.113 with $p = 0.001$. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall

et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 5.310 and standard deviation 17.55 in the local panel, while the main regressor Breadth of recombination has mean 4.294 and standard deviation 4.567 (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.464 to 8.115 while the mean regressor ranges from 2.411 to 9.963. This does not substitute for the model, but it shows how much raw variation the specification must explain (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005).

6.4 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within rosenkopf_impact, the leading coefficient family reads as follows: Boundary new field versus local = 0.113 (p = 0.001); Recombination degree = -0.411 (p = 0.023); total_patents_same_year = -7.48e-05 (p = 0.036); Prior-art citations = 0.0013 (p = 0.055); Log prior patent stock = -0.196 (p = 0.066). The largest positive estimate in this specification is Boundary new field versus local at 0.113, while the largest negative estimate is Recombination degree at -0.411. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert,

2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within rosenkopf_top5, the leading coefficient family reads as follows: Log prior patent stock = -0.421 (p = 0.018); Boundary existing field versus local = -0.232 (p = 0.15); Prior-art citations = 0.0055 (p = 0.17); total_patents_same_year = -5.93e-05 (p = 0.34); Boundary new field versus local = 0.104 (p = 0.36). The largest positive estimate in this specification is Boundary new field versus local at 0.104, while the largest negative estimate is Log prior patent stock at -0.421. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Rosenkopf and Nerkar (2001) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Rosenkopf and Nerkar, 2001; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.5 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Rosenkopf Impact, the most informative focal term is Boundary new field versus local with estimate 0.113 and p = 0.001; In Rosenkopf Top5, the most informative focal term is Log prior patent stock with estimate -0.421 and p = 0.018 (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The Rosenkopf Impact specification contributes its own reading to the replication. Boundary new field versus local is estimated at 0.113 with p = 0.001 and is classified as a strong signal; Recombination degree is estimated at -0.411 with p = 0.023 and is classified as a moderate signal; Prior-art citations is estimated at 0.0013 with p = 0.055 and is classified as a suggestive signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and

March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The Rosenkopf Top5 specification contributes its own reading to the replication. Log prior patent stock is estimated at -0.421 with p = 0.018 and is classified as a moderate signal; Boundary existing field versus local is estimated at -0.232 with p = 0.15 and is classified as a weak signal; Prior-art citations is estimated at 0.0055 with p = 0.17 and is classified as a weak signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). This model comparison matters for Rosenkopf and Nerkar (2001) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002).

6.6 Diagnostic Evidence

A separate model-fit export is not available for this paper, so the interpretation relies on the coefficient file and the observed distribution of the current dataset (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central claim, measurement architecture, sample definition, estimator, effect pattern, and substantive conclusion are set alongside the current local replication so the reader can see where agreement is real and where it is conditional on design translation. (Rosenkopf and Nerkar, 2001). For Rosenkopf and Nerkar (2001), the published paper points readers toward the following result logic: The pub-

lished paper reports that boundary-spanning exploration can outperform local search, especially when search reaches genuinely new knowledge domains. The local rerun instead yields the following focal evidence pattern: Boundary Existing Field vs Local = -0.0389 with p = 0.39; Boundary New Field Versus Local = 0.113 (p = 0.001); Log Prior Patent Stock = -0.421 (p = 0.018); Recombination Degree = -0.411 (p = 0.023); Prior-Art Citations = 0.0013 (p = 0.055). Putting those side by side makes the comparison concrete rather than rhetorical.

Aspect	OriginalPaper	LocalDataset	Reading
Research question and mechanism	The source paper asks whether boundary-spanning search improves inventive impact relative to local search.	The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel.	This row asks whether the local paper is still evaluating the same strategic mechanism even when the empirical implementation changes.
Data and measurement architecture	The published design distinguishes local search from boundary-spanning search and emphasizes impact rather than raw patent volume.	Patent-level recombination panel built from the cleaned reproduction dataset. Search type reconstructed from prior organizational and field experience counts. Impact outcomes measured with patent-level inventive impact and a top-5-percent indicator.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the substantive target.

Aspect	OriginalPaper	LocalDataset	Reading
Sample, unit, and window	The published study is a patent-based archival analysis with industry-specific coding of exploratory search.	Patent-level recombination panel derived from the cleaned local reproduction dataset. 23,844 observations across 184 entities.	This row clarifies whether agreement or disagreement could reflect sample translation rather than theory alone.
Estimator and identifying comparison	patent-level impact models that compare boundary-spanning search modes with local search while preserving an industry-specific exploratory-search taxonomy The key published comparison asks whether the more exploratory search modes, especially exploration into genuinely new fields, outperform local search in terms of inventive impact rather than patenting volume	Patent-level impact and top-tail models with search-type indicators and patent-level controls Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search	This row separates estimator translation from theory translation.

Aspect	OriginalPaper	LocalDataset	Reading
Effect pattern in the evidence	The published paper reports that boundary-spanning exploration can outperform local search, especially when search reaches genuinely new knowledge domains.	Boundary Existing Field vs Local = -0.0389 with p = 0.39;	
allowbreak Boundary New Field Versus Local = 0.113 (p = 0.001); allowbreak Log Prior Patent Stock = -0.421 (p = 0.018); allowbreak Recombination Degree = -0.411 (p = 0.023);			
allowbreak Prior-Art Citations = 0.0013 (p = 0.055)	This row compares the original paper's claimed effect structure to the actual local estimates.		

Aspect	OriginalPaper	LocalDataset	Reading
Conclusion drawn from the evidence	Boundary-spanning exploration, especially into genuinely new knowledge fields, yields higher inventive impact than local search.	The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Boundary New Field Versus Local = 0.113 (p = 0.001);	
allowbreak Log Prior Patent Stock = -0.421 (p = 0.018). The conclusion is therefore mixed rather than empty.	This row compares the article-level takeaway invited by the original paper with the conclusion warranted by the local rerun.		

Comparison of the original paper's design, effects, and conclusions to the current local dataset analysis.

Effect	Estimate	P-value	Reading
Boundary New Field Versus Local	0.113	0.001	positive and statistically distinguishable in the local replication
Log Prior Patent Stock	-0.421	0.018	negative and statistically distinguishable in the local replication
Recombination Degree	-0.411	0.023	negative and statistically distinguishable in the local replication

Effect	Estimate	P-value	Reading
Prior-Art Citations	0.0013	0.055	positive and only marginally distinguishable in the local replication

Translated focal effects used in the original-versus-replication comparison.

7.2 Detailed Effect and Conclusion Comparison

At the level of article content, the original paper contributes the following substantive message: The published paper reports that boundary-spanning exploration can outperform local search, especially when search reaches genuinely new knowledge domains. The local replication studies the same underlying mechanism through the current executable design: The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. This is why the comparison has to extend beyond a headline coefficient (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). At the level of effects, the original article emphasizes the following empirical pattern: The published paper reports that boundary-spanning exploration can outperform local search, especially when search reaches genuinely new knowledge domains. The current local replication instead shows the following paper-specific focal estimates: Boundary New Field Versus Local = 0.113 (p = 0.001); Log Prior Patent Stock = -0.421 (p = 0.018); Recombination Degree = -0.411 (p = 0.023); Prior-Art Citations = 0.0013 (p = 0.055). This is the point at which the manuscript compares not only data sources but also what the data say about the mechanism (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). At the level of conclusion, the original article leads to the following takeaway: Boundary-spanning exploration, especially into genuinely new knowledge fields, yields higher inventive impact than local search. The current replication instead supports the following more careful conclusion: The local replication does not recover the headline

term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Boundary New Field Versus Local = 0.113 (p = 0.001); Log Prior Patent Stock = -0.421 (p = 0.018). The conclusion is therefore mixed rather than empty (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). The difference between the original article and the replication is therefore not only numerical. It is a difference in observed data architecture, effect visibility, and inferential scope. The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002).

7.3 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design distinguishes local search from boundary-spanning search and emphasizes impact rather than raw patent volume By contrast, the local paper relies on the following executable design: The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published study is a patent-based archival analysis with industry-specific coding of exploratory search The local sample instead works from Patent-level recombination panel derived from the cleaned local reproduction dataset and yields 23,844 observations across 184 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable

ways. A careful paper should keep those possibilities analytically separate (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). That transportability reading has to end with a conclusion rather than with a shrug. In the current local rerun, the conclusion is: The local replication does not recover the headline term cleanly, but the broader focal effect set still contains statistically distinguishable structure, especially Boundary New Field Versus Local = 0.113 (p = 0.001); Log Prior Patent Stock = -0.421 (p = 0.018). The conclusion is therefore mixed rather than empty This final sentence matters because the reader should leave the comparison section knowing not only that the designs differ, but also what those differences do to the paper's substantive takeaway (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as

methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002).

8 Discussion

The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The local search-type coding generalizes the original industry-specific implementation (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. Mismatch with published estimates can arise from both sample expansion and measurement differences (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The exercise is most informative about directional mechanisms (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search (Rosenkopf and Nerkar, 2001; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001;

Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8,443	5.860	4.58%
local	12,131	5.136	4.11%
boundary_existing_field	3,270	4.535	2.87%

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	23,844	5.310	17.55	1.000	2.000	6.000	0	812
Top-5 indicator	23,844	0.0411	0.199	0.0000	0.0000	0.0000	0	1
Breadth of recombination	23,844	4.294	4.567	2.000	3.000	5.000	2	117
Recombination degree	23,844	0.555	0.174	0.444	0.560	0.691	0	1
Field experience	23,844	1.634	1.820	0.0000	1.099	2.773	0	8
Organizational experience	23,844	0.923	1.317	0.0000	0.693	1.386	0	8
Prior-art citations	23,844	15.09	50.45	5.000	8.000	13.00	2	2263
Same-year patents	23,844	728.7	1,303	109.0	313.0	652.0	1	7321
Prior cumulative stock	23,844	7.507	2.214	6.330	7.693	8.893	0	11

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Inventive impact	Top-5 indicator	Breadth of recombination	Recombination degree	Field experience	Organizational experience	Prior-art citations	Same-year patents	Prior cumulative stock
Inventive impact	1.000	0.455	0.364	0.018	-0.021	-0.008	0.591	0	0.022
Top-5 indicator	0.455	1.000	0.129	0.024	-0.012	0.006	0.149	0	0.005
Breadth of recombination	0.364	0.129	1.000	0.466	-0.263	-0.144	0.822	0	0.149
Recombination degree	0.018	0.024	0.466	1.000	-0.456	-0.313	0.140	0	0.122
Field experience	-0.021	-0.012	-0.263	-0.456	1.000	0.838	-0.077	0	0.132
Organizational experience	-0.008	0.006	-0.144	-0.313	0.838	1.000	-0.018	0	0.267
Prior-art citations	0.591	0.149	0.822	0.140	-0.077	-0.018	1.000	0	0.081
Same-year patents	-0.018	-0.015	0.011	0.065	0.060	0.152	-0.016	1	0.520
Prior cumulative stock	0.022	0.005	0.149	0.122	0.132	0.267	0.081	1	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	MeanImpact	MeanBreadth	MeanDegree	NewFieldShare
1986	643	4.020	2.869	0.517	42.77%
1987	789	4.280	2.918	0.518	43.35%
1988	1,066	4.461	3.132	0.532	43.53%
1989	1,390	3.934	3.081	0.527	40.07%
1990	1,349	4.514	3.116	0.522	37.44%
1991	1,325	4.738	3.150	0.531	37.43%
1992	1,302	5.015	3.332	0.535	40.40%
1993	1,354	5.672	3.461	0.532	38.77%
1994	1,330	6.265	3.578	0.541	39.32%
1995	1,217	6.260	3.844	0.556	37.80%
1996	659	6.727	3.838	0.531	37.63%
1997	646	6.014	4.245	0.552	41.95%
1998	356	6.295	3.924	0.547	40.45%
1999	654	5.483	3.810	0.544	37.00%
2000	636	5.541	4.022	0.540	36.32%
2001	614	5.151	4.414	0.562	41.69%
2002	632	4.889	4.384	0.558	36.87%
2003	522	4.502	4.418	0.550	34.87%
2004	324	3.664	4.809	0.573	41.36%
2005	284	4.500	5.486	0.576	40.85%
2006	336	4.890	5.265	0.566	32.14%
2007	246	3.654	5.268	0.569	30.89%
2008	535	3.652	4.512	0.590	36.07%
2009	206	4.801	6.005	0.599	29.61%
2010	511	4.417	6.648	0.608	25.64%
2011	651	6.452	6.275	0.609	25.50%
2012	1,068	4.000	6.519	0.611	26.03%
2013	868	5.386	6.730	0.598	26.04%

Period	Patents	MeanImpact	MeanBreadth	MeanDegree	NewFieldShare
2014	725	12.64	6.989	0.600	22.90%
2015	1,072	6.616	6.174	0.597	20.43%
2016	534	3.768	4.629	0.581	16.67%

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	13,675	2.411	4.464
Q2	3,458	4.000	4.925
Q3	2,273	5.000	5.505
Q4	4,438	9.963	8.115

Quantile summary for the main regressor.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8,443	5.860	4.58%
local	12,131	5.136	4.11%
boundary_existing_field	3,270	4.535	2.87%

Raw group means from the local summary file.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value	model
search_type:: allowbreakboundary_existing_field	-0.0389	0.0449	-0.866	0.39	rosenkopf_impact
search_type:: allowbreakboundary_new_field	0.1126	0.0353	3.188	0.001	rosenkopf_impact
breadth_recombination	0.0223	0.0130	1.711	0.087	rosenkopf_impact
recombination_degree	-0.4108	0.1802	-2.280	0.023	rosenkopf_impact
n_prior_art_cites	0.0013	0.0007	1.921	0.055	rosenkopf_impact
total_patents_same_year	-7.48e-05	3.56e-05	-2.099	0.036	rosenkopf_impact
total_patents_until_prev_log	-0.1965	0.1067	-1.841	0.066	rosenkopf_impact
search_type:: allowbreakboundary_existing_field	-0.2317	0.1623	-1.428	0.15	rosenkopf_top5
search_type:: allowbreakboundary_new_field	0.1039	0.1131	0.918	0.36	rosenkopf_top5
breadth_recombination	0.0069	0.0205	0.339	0.73	rosenkopf_top5
recombination_degree	-0.0597	0.2786	-0.214	0.83	rosenkopf_top5
n_prior_art_cites	0.0055	0.0041	1.358	0.17	rosenkopf_top5
total_patents_same_year	-5.93e-05	6.16e-05	-0.962	0.34	rosenkopf_top5

term	estimate	std_error	statistic	p_value	model
total_patents_until_prev_log	-0.4208	0.1786	-2.357	0.018	rosenkopf_top5

Full coefficient dump exported from the local estimation script.

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